A-A-53140A
25 January 1988
Superceding
A-A-0053140(DM)
18 August 1987

COMMERCIAL ITEM DESCRIPTION

COMPARATOR SET, COLOR (Free Residual Chlorine and pH Determination)

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers an indicator kit for use in performing pH and free residual chlorine determinations of drinking water in the field.

Salient characteristics:

Shall be an indicator kit for use in performing pH and free residual chlorine determinations of drinking water in the field.

Shall consist of the following components:

1. A single color comparator with prismatic eyepiece assembly that uses interchangeable color disks for making the pH and free chlorine determinations.

2. A chlorine color disk for use by the DPD method in determining free chlorine in concentrations ranging from 0.1 to 10 parts per million (ppm) chlorine, in nine (9) step graduations 0.1, 0.4, 0.7, 1.0, 1.5, 2.0, 3.0, 5.0 and 10.0 ppm.

3. A pH color disk for use with wide range pH indicator solutions in determining pH in the range of 3.0 to 11.0 pH, in nine (9) step graduations of 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0 and 11.0 pH.

4. Four (4) rectangular sample cells fabricated of colorless and optically clear glass. Each sample cell shall have nominal millimeter (mm) inside dimensions of 26 depth X 13 width X 77 length, and a bottom thickness of 1.5 to 3.5 mm. Each sample cell shall have a 15 milliliter (ml) capacity, and shall include a graduation mark to designate the 15 ml capacity level. The numerical value shall be included above the 15 ml graduation mark.

AMSC N/A FSC 6630

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5. One 1/2 ounce capacity pH reagent bottle (empty) fabricated of amber colored glass and supplied in the form of a French square narrow mouth bottle. The bottle shall be labeled to indicate its intended use with a wide range pH reagent. The bottle shall be supplied with a rubber bulb dropping pipette that is graduated and numbered to indicate the 0.5 ml capacity level.

6. A carrying case with operating instructions.

7. Service data.

Shall have chlorine and pH color disks fabricated of suitable black plastic. Each disk shall contain nine (9) calibrated, permanent, nonfading color glass standards. The color glass standards shall be evenly spaced and positioned near the perimeter of each disk. Corresponding numerical values for the calibrated color glass standards shall be permanently marked in black print on securely affixed circular inserts.

Shall have a comparator housing that is fabricated from an impact resistant black plastic. The comparator housing shall be constructed to include the following:

1. A back section that contains an attached opal glass screen. The opal glass screen shall be attached to the housing in such a manner that the comparator can be used with natural light or by holding a standard battery operated flashlight against the screen. The opal glass screen selected shall provide an even diffusion of light over both a test sample and a color standard on an inserted disk.

2. Compartments for holding two (2) inserted sample cells in upright positions. One compartment shall accomodate a sample cell with test water that has been treated with a pH reagent (for pH determinations) or with two crushed DPD No. 1 tablets (for chlorine determinations). The other compartment shall accomodate the other sample cell with untreated test water (i.e. a control sample) that shall be positioned behind a rotating disk and its color standards.

3. A compartment for inserting a color disk in such a manner that it partially protrudes from the housing to facilitate fingertip control (for rotation). An inserted disk shall be held in place without any tendency to shift out of position when being rotated.

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4. A front face that includes a prismatic eyepiece assembly and a suitable cutout for checking the numerical value of a color disk standard that is being seen through the prismatic eyepiece.

Shall have the optics of the prismatic eyepiece assembly fabricated from borosilicate glass or methyl methacrylate. The prismatic optics shall be designed to bring the view of a color glass standard into juxtaposition with that of sample cell containing a treated water sample.

Shall be supplied with a carrying case with hinged top, latch and carrying handle. The case shall include compartments for holding the comparator, pH and chlorine color disks and pH reagent bottle. The carrying case shall be constructed from materials suitable to withstand impact. The inner side of the carrying case cover shall include brief operating instructions for the comparator. The operating instructions shall be printed on a paper card that is covered on both sides by being laminated on both sides with clear plastic.

Workmanship. The indicator kit shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One comparator set, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purhase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of ~~special laboratory appliances or procedures) and testing~~ (determination by technical means of physical and chemical properties) of the item.

Inspections and tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements.

Sampling for inspections and tests. Sampling for inspections and tests shall be conducted in accordance with commercial practice.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

6630-01-044-0334

Ordering data. (Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

AGENT:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS

Project No. 6630-0464

Location: Disk # 10

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53140A	COMPARATOR SET, COLOR (Free Residual Chlorine and pH Determinatio

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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